INTERNSHIP REPORT
ON
Swiftpay –
International Remittance & Currency Exchange Portal.

Internship Report submitted
to
Shri Ramdeobaba College of Engineering & Management, Nagpur in partial fulfillment of requirement for the award of degree of

Bachelor of Technology

In

Computer Science and Engineering

By

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CERTIFICATE

This is to certify that the Internship Report on **Swiftpay – International Remittance and Currency Exchange Portal** is a bonafide work of Sohaib Munir Shaikh submitted to the Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur in partial fulfillment of the award of a Bachelor of Engineering, in Computer Science and Engineering has been carried out at the **Cognizant Technology Solutions, Chennai** during the academic year 2025-2026.

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I, hereby declare that the Project titled **Skillforge – International Remittance and Currency Exchange Portal** submitted herein, has been carried out in the **Cognizant Technology Solutions, Chennai**. The work is original and has not been submitted earlier as a whole or part for the award of any degree / diploma at this or any other institution / University.

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- Sohaib Munir Shaikh

ABSTRACT

This report presents the work carried out during an approximately 15-week internship, focused on the design and development of a web-based application titled *Swiftpay – International Remittance and Currency Exchange Portal*. The internship primarily emphasized full-stack development using the .NET framework for backend services and React for frontend development, along with the application of database management concepts and RESTful API integration.

The internship was structured to provide a comprehensive understanding of enterprise-level application development, including system design, modular architecture, role-based access control, and scalable web application practices. During the development process, foundational concepts in backend development using ASP.NET Core, frontend development using React, and relational database management were strengthened. Practical exposure was gained in building responsive user interfaces, implementing business logic, handling data persistence, and ensuring secure and efficient communication between system components.

The major project undertaken during the internship was the development of the Swiftpay platform, designed to manage workforce training, certification processes, and competency tracking within an organization. The system incorporated multiple modules such as Identity and Access Management, Training Catalog and Course Management, Enrollment and Attendance Tracking, Assessment and Certification Management, Competency and Skill Gap Analysis, Compliance and Audit Management, and Reporting and Analytics. The application supported multiple user roles including employees, trainers, managers, HR officers, and administrators, each with specific functionalities and access privileges.

The system was developed using a microservices-inspired architecture with REST APIs, ensuring modularity, scalability, and maintainability. Key functionalities included course creation and enrollment workflows, assessment handling, certification issuance, performance tracking, and automated notifications. The application also emphasized data-driven decision-making through dashboards and reporting features.

This internship significantly enhanced technical proficiency in full-stack development, system design, API integration, and database management. Additionally, it improved problem-solving abilities, debugging skills, and understanding of real-world enterprise application workflows. The experience also contributed to better documentation practices and an overall understanding of scalable software development in a professional environment.

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CHAPTER 1 INTRODUCTION

Internship training is an essential part of engineering education because it bridges the gap between academic learning and industry practices. It gives students the opportunity to understand how concepts learned in classrooms are applied in real software projects, team environments, and business-driven workflows.

Fig. 1.1: Cognizant Logo

As part of the internship program spanning approximately 15 weeks, the role undertaken involved working on fullstack web application development using .NET and React. The internship focused on understanding enterprise application architecture, RESTful API design, and the implementation of scalable and modular systems for real-world business use cases.

The work carried out during this internship revolved around the development of the *SkillForge – Workforce Training and Certification Management System*, a web-based platform designed to streamline training management, certification tracking, and competency analysis within organizations. The development process involved backend implementation using ASP.NET Core, frontend development using React, and database management using a relational database system.

The internship also involved designing and implementing multiple functional modules such as user authentication and role-based access control, course and training management, enrollment tracking, assessment handling, certification management, and reporting dashboards. These modules were integrated through REST APIs to ensure efficient communication between system components and to maintain a modular and scalable architecture.

This internship experience contributed significantly to building a strong foundation in full-stack development, system design, and real-world application workflows. It also enhanced problem-solving skills, debugging capabilities, and the ability to develop structured, maintainable, and user-centric software solutions in a collaborative development environment.

CHAPTER 2 TOOLS AND TECHNOLOGIES

During the internship, a diverse set of technologies was utilized to support the design and development of a full-stack web application. These tools collectively enabled the implementation of scalable backend services, responsive user interfaces, and efficient data management.

2.1 .NET

.NET was used as the primary framework for backend development. It enabled the implementation of robust and scalable server-side logic using ASP.NET Core, supporting the creation of RESTful APIs and modular application architecture. The framework facilitated the use of object-oriented programming principles, efficient exception handling, and structured code organization through controllers, services, and models. .NET also played a key role in integrating database operations through Entity Framework Core, enabling seamless data access, and supporting the overall development of a maintainable and enterprise-level web application.

Fig. 2.1: .NET Logo

2.2 Entity Framework Core for .NET Core

Entity Framework Core served as the primary data access technology used during the internship. It enabled efficient interaction with the database through object-relational mapping, allowing application data to be handled using domain models instead of complex SQL queries. Entity Framework Core was used for performing CRUD operations, managing relationships between entities, and ensuring data consistency within the application. It also supported features such as migrations, query handling using LINQ, and change tracking, which simplified database management and improved overall development efficiency.

Fig. 2.2: Entity Framework Core Logo

2.3 SQL

SQL was learned and practiced to understand how structured data is stored, queried, and validated. It was useful in developing logical thinking for data verification and strengthened the ability to work with test data, filtering operations, and backend validation concepts relevant to enterprise testing.

Fig. 2.3: SQL Logo

2.4 React JS

React was used as the primary frontend framework for building dynamic and responsive user interfaces. It enabled the development of interactive web pages through component-based architecture, data binding, and modular design. React facilitated the implementation of user interactions such as form handling, data display, navigation, and real-time updates. It was especially useful in structuring the application into reusable components, managing client-side logic, and ensuring seamless communication with backend APIs, resulting in a smooth and efficient user experience.

Fig. 2.4:React JS Logo

2.5 Postman

Postman was used for understanding and manually validating APIs. It provided experience in sending HTTP requests, analyzing JSON responses, checking status codes, and understanding authentication, parameters, headers, and payload structures before moving toward automated API testing.

Fig. 2.5: Postman Logo

2.6 Swagger

Swagger was used as an API documentation and testing tool during the development process. It provided an interactive interface to visualize and test RESTful APIs without the need for external tools. Swagger enabled easy verification of API endpoints, request and response formats, and parameter handling. It was particularly useful for debugging, validating backend functionality, and ensuring smooth communication between frontend and backend components.

Fig. 2.6: Swagger Logo

Technology Summary

Technology	**Purpose in Internship**
C# / .NET (ASP.NET Core)	Core backend development and REST API implementation
React	Frontend development for dynamic and responsive UI

SQL	Data querying and understanding of backend validation
Entity Framework Core	Database interaction and ORM-based data management
Postman	Manual API testing and request validation
Swagger	API documentation and testing

Table 2.1: Technology Stack Used During Internship

CHAPTER 3 INTERNSHIP DOMAIN

3.1. Overview of the Domain

The internship domain was **Full-Stack Web Application Development**. This domain focuses on designing and developing complete web-based systems that integrate user interfaces, backend services, and database management to deliver functional, scalable, and user-centric applications.

Full-stack development is not limited to building interfaces or writing server-side code. It involves designing system architecture, implementing business logic, managing data flow, ensuring secure communication through APIs, and maintaining application performance and reliability. In modern development environments, it emphasizes modular design, reusability, and efficient handling of realworld workflows.

During the internship, this domain was explored through the development of a web-based platform using .NET and React. The work performed reflected how full-stack development supports features such as user authentication, role-based access, course and training management, data processing, and reporting within an enterprise application.

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3.2. Relevance of the Domain

The relevance of the Full-Stack Web Development domain is very high in the software industry, as modern applications rely on seamless integration of frontend, backend, and database systems. The performance, usability, and reliability of web applications directly impact user experience, business efficiency, and organizational success. Any failure in core functionalities such as authentication, data processing, or system workflows can lead to operational issues and reduced user trust.

The skills covered in the internship (Dotnet(C#) , Entity Framework , React , API testing) are highly relevant across multiple sectors:

- **Enterprise Systems :** Managing internal operations such as training, customer management.
- **Finance:** Enabling secure, data-driven applications for transaction processing and reporting.

This domain plays a critical role in building scalable and efficient digital solutions that support real-world business processes across industries.

3.3. Problem Areas in the Domain

Common problem areas in the Full-Stack Web Development domain include inefficient system architecture, poor integration between frontend and backend components, data inconsistency, and performance bottlenecks. Issues such as improper API design, slow response times, and inadequate error handling can negatively impact application reliability and user experience. Additionally, managing secure authentication, role-based access, and handling large volumes of data across multiple modules presents significant challenges.

Handling diverse data and ensuring smooth communication between system components requires well-structured APIs and efficient data management strategies. As applications grow, maintaining scalability, modularity, and code maintainability becomes increasingly complex.

These challenges are addressed through structured development practices such as:

- **Modular Architecture :** Designing independent and reusable components for better maintainability.
- **Efficient API Design :** Ensuring reliable and secure communication between frontend and backend systems.
- **Scalability:** Building systems that can handle increasing users and data without performance degradation.
- **Data Management:** Implementing optimized database operations and consistency mechanisms.

3.4. Importance of Data in the Domain

Data plays a critical role in Full-Stack Web Development, as it drives application functionality, decision-making, and user interactions. Efficient data handling ensures that information is accurately stored, processed, and retrieved across different modules of the system.

In web applications, data is essential for managing user information, handling business workflows, supporting authentication and authorization, and generating reports. Proper data management enables smooth communication between frontend and backend components through APIs, ensuring consistency and reliability of the application.

Additionally, handling large volumes of data, maintaining data integrity, and ensuring secure access are key challenges in this domain. Effective data management practices, including structured database design, validation mechanisms, and optimized queries, are crucial for building scalable and high-performance applications.
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3.5. Key Learnings from the Domain

The internship in Full-Stack Web Development provided a comprehensive understanding of application development practices, combining both theoretical concepts and practical implementation. It helped build a strong foundation in designing and developing web applications while demonstrating how different system components integrate in real-world scenarios.

The experience offered hands-on exposure to frontend and backend development, API integration, database management, and modular architecture. It also highlighted the importance of writing maintainable code, handling data efficiently, and ensuring seamless communication between application layers.

Overall, the internship enhanced technical skills, problem-solving abilities, and understanding of enterpriselevel development workflows, contributing significantly to practical knowledge and professional growth.

CHAPTER 4 WORKING METHODOLOGY

The internship followed a structured methodology over approximately 15 weeks, where learning, design, and implementation were aligned progressively. The workflow began with understanding fundamental concepts of full-stack development and gradually transitioned into practical application through project development.

4.1 Foundation Building

During the initial phase of the internship, the learning progressed in a structured manner starting from backend development fundamentals to frontend development and database handling. The journey began with C# and .NET concepts, followed by database management using SQL, and then frontend development using React. This stage involved:

- **C# and .NET Fundamentals:** A strong foundation was built in C# and ASP.NET Core, covering concepts such as object-oriented programming, controllers, services, middleware, and exception handling, which are essential for developing scalable backend applications.

- **Learning SQL and Entity Framework** : Proficiency was developed in writing queries for data extraction, filtering, joins, and aggregations, which helped in managing and validating application data effectively.

- **React Basics**: Frontend development skills were developed using React, including component-based architecture, data binding, routing, and form handling for building interactive user interfaces.

- **Hands-On Practice**: Practical exercises combined backend development, database operations, and frontend integration to simulate real-world application development scenarios and strengthen full-stack development skills.

4.2 Tool Familiarization and Practice

In the second stage of the internship, the focus shifted towards hands-on application of development tools and frameworks, along with exposure to Agile-based development practices. This phase emphasized practical learning through guided implementation, helping to understand how different technologies integrate into a full-stack application workflow.

The development approach followed Agile methodology principles, where work was carried out in iterative cycles with continuous improvement, regular testing, and incremental feature development. This helped in better task management, faster issue resolution, and maintaining flexibility during development.

- **ASP.NET Core:** Gained experience in building RESTful APIs, implementing controllers, services, and middleware, and handling business logic for backend development.

- **Entity Framework Core:** Learned to interact with the database using ORM techniques, perform CRUD operations, manage relationships, and handle data migrations efficiently.

- **React:** Developed frontend skills by working with components, data binding, routing, and form handling to create dynamic and responsive user interfaces.

- **Swagger & Postman:** Used for testing and validating API endpoints, understanding request-response cycles, and ensuring smooth communication between frontend and backend components.

- **Version Control (Git):** Practiced managing code changes, maintaining version history, and supporting iterative development in an Agile environment.

- **Agile Methodology:** Followed an iterative development approach with modular implementation, continuous testing, and regular refinement of features, ensuring steady progress and adaptability during the project lifecycle.

4.3 Project Development

After completing the foundational learning, the focus shifted to the development of the major project, *Swiftpay – International Remittance and Currency Exchange Portal*. This phase emphasized applying full-stack development concepts in a real-world scenario, covering the complete application lifecycle from planning to implementation.

- **System Design & Planning:** Designed the overall system architecture by identifying key modules such as user authentication, course management, enrollment, assessments, certification, and reporting. Defined workflows for different user roles including customers, agents, treasury, operations, and administrators.
- **Backend Development:** Implemented RESTful APIs using ASP.NET Core to handle business logic, user management, data processing, and secure communication between components. Structured the application using controllers, services, and models for better maintainability.

- **Frontend Development:** Developed dynamic and responsive user interfaces using React, including form handling, data binding, routing, and component-based design to support various application features.

- **Database Integration:** Used Entity Framework Core to manage database operations such as CRUD functionalities, relationships between entities, and data consistency across modules.

- **API Integration & Validation:** Ensured seamless communication between frontend and backend through REST APIs, validating request-response cycles and handling errors effectively.

- **Testing & Debugging:** Performed continuous testing and debugging to ensure correct functionality of modules, smooth user workflows, and reliable system performance.

The final stage of the project focused on integrating all modules, refining system performance, and ensuring end-to-end workflow execution across different user roles. This phase highlighted the importance of scalability, modular design, and maintainable code in enterprise-level application developmen

4.4 Week-wise Breakdown

Week Range	**Focus Area**	**Outcomes**
Week 1	C# and .NET Fundamentals	Built strong programming foundation and understanding of backend concepts
Week 2	Project Initiation & System Design	Defined modules, user roles, workflows, and overall system architecture
Week 3–4	Backend Development (API Creation)	Developed REST APIs using ASP.NET Core for core functionalities
Week 5–6	Database Design & Integration	Designed schema, implemented queries, and connected APIs with database
Week 7–8	Advanced Backend Development	Implemented business logic, authentication, and role-based access control
Week 9–10	React Development	Built frontend components, routing, and user interface features

Week 11–12	API Integration (Frontend + Backend)	Connected React frontend with backend APIs and handled data flow
Week 13–14	Module Implementation (SkillForge)	Developed modules like course management, enrollment, and reporting
Week 15	Testing, Debugging & Final Integration	Performed end-to-end testing, debugging, and system refinement

Table 4.1: Week-wise Learning and Work Breakdown

CHAPTER 5 PROJECT FLOW AND ARCHITECTURE

The *Swiftpay – International Remittance and Currency Exchange Portal* was developed using a layered and modular architecture to ensure scalability, maintainability, and efficient data flow between system components. The application follows a full-stack approach, integrating a frontend user interface, backend services, and a relational database.

5.1 Project Flow:

The overall workflow of the system is designed to support multiple user roles and their interactions within the platform. The general flow of the application is as follows:

- **User Authentication & KYC Onboarding:** Users (Senders) log into the system using secure credentials. New users must complete **Know Your Customer (KYC)** verification by uploading identification documents, which are verified by the system or compliance officers before any transactions are permitted.

- **Dashboard & Wallet Overview:** After login, users access a dashboard showing their transaction history, saved recipients, current exchange rates, and multi-currency wallet balances. Admins and Agents see high-level liquidity and pending approval queues.

- **Transaction Initiation:** The Sender initiates a transfer by selecting a recipient, entering the amount, and choosing the payout method (Bank Transfer, Mobile Money, or Cash Pickup). The system provides a real-time breakdown of exchange rates, service fees, and the final amount to be received.

- **Compliance & Fraud Screening:** Every transaction undergoes automated **Anti-Money Laundering (AML)** and **Counter-Terrorist Financing (CTF)** screening. Transactions that trigger red flags are routed to Compliance Officers for manual review.

- **Payment Processing & Settlement:** Once cleared, the sender pays for the transaction via integrated gateways (Credit/Debit card, ACH, or local bank transfer). The system then interfaces with liquidity providers or local partner banks to execute the payout in the destination country.

- **Payout & Recipient Notification:** The system generates a unique transaction reference number. The Recipient receives the funds or is notified to collect cash. Status updates are pushed to both the sender and recipient in real-time.

- **Reporting & Reconciliation:** Admins and HR/Finance personnel monitor daily volumes, track settlement balances with foreign partners, and generate regulatory reports for central banks or financial authorities.

- **Support & Dispute Management:** The system provides a dedicated channel for users to raise tickets for delayed transfers or failed payments, allowing Support Staff to investigate and trigger refunds if necessary.

5.2 System Architecture:

The application follows a **three-tier architecture** consisting of the following layers:

- **Frontend Layer (Presentation Layer):** Developed using React, this layer handles user interaction, UI rendering, form inputs, and data presentation. It communicates with the backend through REST APIs.

- **Backend Layer (Application Layer):** Built using ASP.NET Core, this layer contains business logic, API endpoints, authentication mechanisms, and role-based access control. It processes client requests and interacts with the database.

- **Database Layer (Data Layer):** A relational database is used to store application data such as user details, courses, enrollments, assessments, and certifications. Entity Framework Core is used for data access and ORM operations.

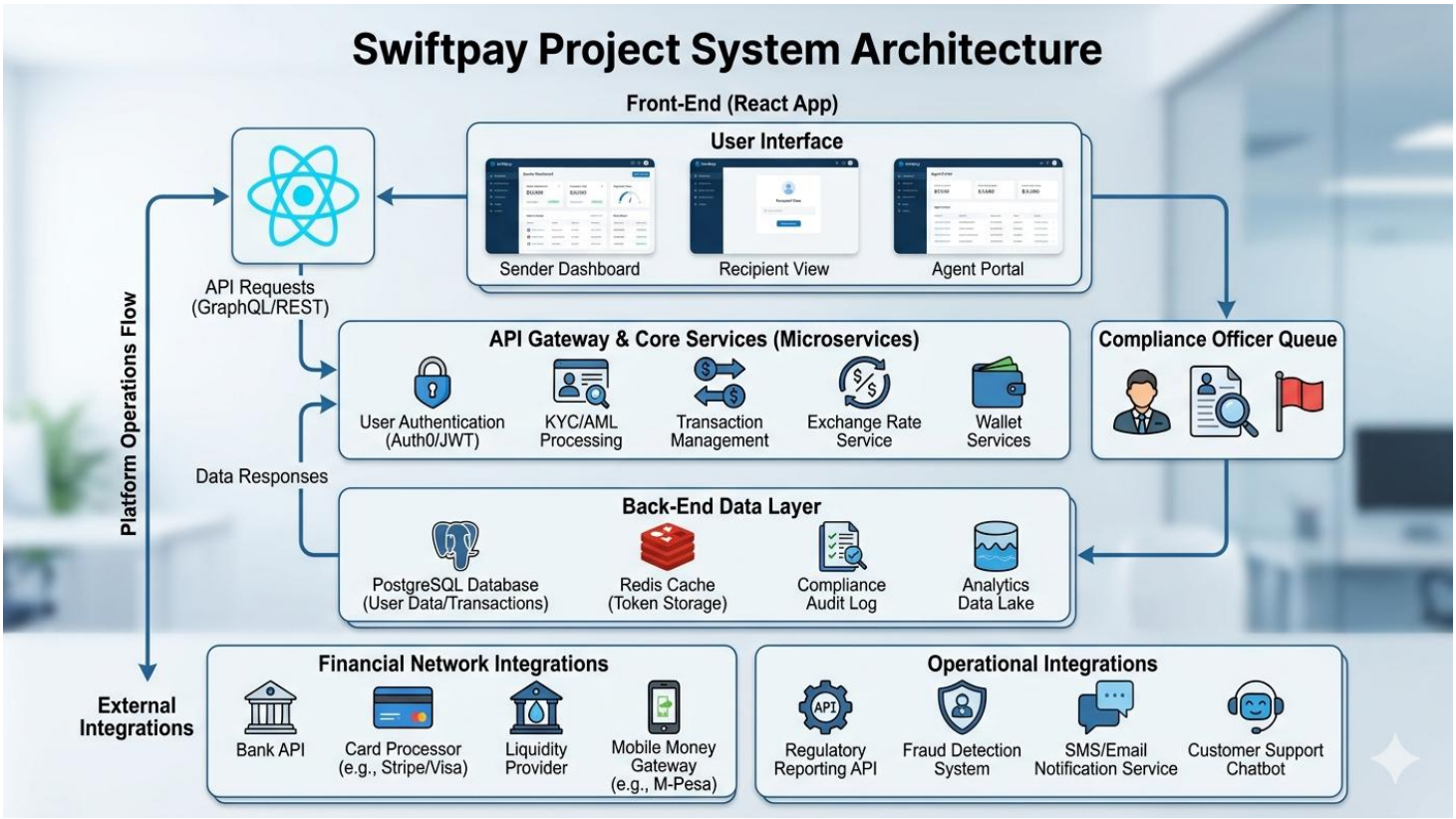

Fig 4.1: System Architecture

5.3 Architectural Features:

- **Modular Design:** Each module (authentication, courses, assessments, etc.) is independently structured for better maintainability.

- **RESTful Communication:** Ensures smooth data exchange between frontend and backend.

- **Scalability:** Designed to handle increasing users and data efficiently.

- **Security:** Implements role-based access control and secure data handling.

- **Maintainability:** Clean separation of concerns between layers improves code management.

CHAPTER 6 ABOUT INTERNSHIP ROLE AND WORK

During my eighth-semester internship, I worked as a Full-Stack Developer Intern in the FullStack Web Development domain, where I was actively involved in the end-to-end development of the *Swiftpay– International Remittance and Currency Exchange Portal*.

The internship spanned approximately 15 weeks and followed an Agile-based development approach, where work progressed iteratively with continuous integration of features and improvements.

The internship began with an onboarding and foundation-building phase, during which I familiarized myself with the project requirements, system design, and development environment. I set up the backend using ASP.NET Core and configured the frontend using React, along with necessary tools such as Entity Framework Core for database operations and Swagger for API testing. This phase also involved understanding coding standards, project structure, and best practices for scalable application development.

In the initial development phase, I focused on backend engineering, where I designed and implemented RESTful APIs for core modules such as user management, course management, enrollment, assessments, and certification. Each API endpoint was structured using controllers, services, and models, ensuring clean separation of concerns. Business logic was implemented within service layers, while database interactions were handled using Entity Framework Core. Proper validation and error handling mechanisms were incorporated to ensure reliability and data integrity.

Following backend development, I worked on frontend implementation using React. I developed reusable components and designed user interfaces for different modules, including dashboards, forms, and data views. Features such as routing, data binding, and form validation were implemented to ensure a smooth and interactive user experience. The frontend was integrated with backend APIs, replacing static data with real-time data exchange and ensuring seamless communication between layers.

In the later stages, the focus shifted to integration, testing, and refinement. I performed end-toend testing of application workflows, identified and resolved bugs, and optimized system performance. The application was structured in a modular and maintainable manner, with clear organization of components, services, and data models. Version control practices were followed using Git to manage code changes and maintain consistency throughout development.

Beyond technical contributions, the internship also provided valuable professional exposure. I gained experience working in an Agile environment, managing tasks in iterative cycles, and continuously improving the application based on feedback.
This helped in developing time management skills, problem-solving abilities, and a disciplined approach to software development.

Key learning outcomes from my role included a strong understanding of full-stack development using .NET and React, practical experience in building scalable web applications, and the ability to handle real-world development challenges. The internship strengthened my confidence in designing, developing, and delivering structured software solutions in a professional environment.

CHAPTER 7 CONCLUSION AND OUTCOME

The internship provided a comprehensive learning experience by bridging the gap between academic knowledge and real-world software development practices. Through the development of the *Swiftpay – International Remittance and Currency Exchange Portal*, practical exposure was gained in designing, developing, and integrating a full-stack web application using .NET and React.

The internship helped in understanding the complete software development lifecycle, including requirement analysis, system design, backend and frontend development, database integration, and testing. Working in an Agile-based environment enabled structured and iterative development, improving adaptability and efficiency in handling tasks and resolving issues.

From a technical perspective, the internship strengthened proficiency in ASP.NET Core, React, Entity Framework Core, REST API development, and database management. It also enhanced the ability to write modular, maintainable, and scalable code while ensuring proper data handling and system performance.

In addition to technical growth, the experience contributed significantly to the development of problem-solving skills, debugging techniques, time management, and professional communication. Exposure to real-world workflows and collaborative development practices helped in building confidence and a disciplined approach toward software engineering.

Overall, the internship was a valuable experience that provided both technical expertise and professional development, preparing for future roles in full-stack web development and enabling meaningful contributions in real-world software projects.